

1. (a)

$$\begin{aligned} x[n] &= \delta[n] \\ X[k] &= \sum_{n=0}^{N-1} \delta[n] W_N^{kn}, \quad 0 \leq k \leq (N-1) \\ &= 1 \end{aligned}$$

(b)

$$\begin{aligned} x[n] &= \delta[n - n_0], \quad 0 \leq n_0 \leq (N-1) \\ X[k] &= \sum_{n=0}^{N-1} \delta[n - n_0] W_N^{kn}, \quad 0 \leq k \leq (N-1) \\ &= W_N^{kn_0} \end{aligned}$$

(c)

$$\begin{aligned} x[n] &= \begin{cases} 1, & n \text{ even} \\ 0, & n \text{ odd} \end{cases} \\ X[k] &= \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad 0 \leq k \leq (N-1) \\ &= \sum_{n=0}^{(N/2)-1} W_N^{2kn} \\ &= \frac{1 - e^{-j2\pi k}}{1 - e^{-j(\pi k/N)}} \quad \text{should be } (4\pi k/N) \\ X[k] &= \begin{cases} N/2, & k = 0, N/2 \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

(d)

$$\begin{aligned} x[n] &= \begin{cases} 1, & 0 \leq n \leq ((N/2) - 1) \\ 0, & N/2 \leq n \leq (N-1) \end{cases} \\ X[k] &= \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad 0 \leq k \leq (N-1) \\ &= \sum_{n=0}^{(N/2)-1} W_N^{kn} \\ &= \frac{1 - e^{-j\pi k}}{1 - e^{-j(2\pi k)/N}} \\ X[k] &= \begin{cases} N/2, & k = 0 \\ \frac{2}{1 - e^{-j(2\pi k)/N}}, & k \text{ odd} \\ 0, & k \text{ even}, 0 \leq k \leq (N-1) \end{cases} \end{aligned}$$

(e)

$$\begin{aligned} x[n] &= \begin{cases} a^n, & 0 \leq n \leq (N-1) \\ 0, & \text{otherwise} \end{cases} \\ X[k] &= \sum_{n=0}^{N-1} a^n W_N^{kn}, \quad 0 \leq k \leq (N-1) \\ &= \frac{1 - a^N e^{-j2\pi k}}{1 - a e^{-j(2\pi k)/N}} \\ X[k] &= \frac{1 - a^N}{1 - a e^{-j(2\pi k)/N}} \end{aligned}$$

2. Consider the finite-length sequence

$$x[n] = \begin{cases} e^{j\omega_0 n}, & 0 \leq n \leq (N-1) \\ 0, & \text{otherwise} \end{cases}$$

(a) The Fourier transform of $x[n]$:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \\ &= \sum_{n=0}^{N-1} e^{j\omega_0 n} e^{-j\omega n} \\ X(e^{j\omega}) &= \frac{1 - e^{-j(\omega - \omega_0)N}}{1 - e^{-j(\omega - \omega_0)}} \\ &= \frac{e^{-j(\omega - \omega_0)(N/2)}}{e^{-j(\omega - \omega_0)/2}} \left(\frac{\sin [(\omega - \omega_0)(N/2)]}{\sin [(\omega - \omega_0)/2]} \right) \\ X(e^{j\omega}) &= e^{-j(\omega - \omega_0)((N-1)/2)} \left(\frac{\sin [(\omega - \omega_0)(N/2)]}{\sin [(\omega - \omega_0)/2]} \right) \end{aligned}$$

(b) N-point DFT:

$$\begin{aligned} X[k] &= \sum_{n=0}^{N-1} x[n]W_N^{kn}, \quad 0 \leq k \leq (N-1) \\ &= \sum_{n=0}^{N-1} e^{j\omega_0 n} W_N^{kn} \\ &= \frac{1 - e^{-j((2\pi k/N) - \omega_0)N}}{1 - e^{-j((2\pi k/N) - \omega_0)}} \\ &= e^{-j(\frac{2\pi k}{N} - \omega_0)(\frac{N-1}{2})} \frac{\sin [(\frac{2\pi k}{N} - \omega_0) \frac{N}{2}]}{\sin [(\frac{2\pi k}{N} - \omega_0)/2]} \end{aligned}$$

Note that $X[k] = X(e^{j\omega})|_{\omega=(2\pi k)/N}$

(c) Suppose $\omega_0 = (2\pi k_0)/N$, where k_0 is an integer:

$$\begin{aligned} X[k] &= \frac{1 - e^{-j(k-k_0)2\pi}}{1 - e^{-j(k-k_0)(2\pi)/N}} \\ &= e^{-j(2\pi/N)(k-k_0)((N-1)/2)} \frac{\sin \pi(k-k_0)}{\sin(\pi(k-k_0)/N)} \end{aligned}$$

3. From Fig P8.10-1, the two 8-pt sequences are related through a circular shift. Specifically,

$$x_2[n] = x_1[((n-4))_8]$$

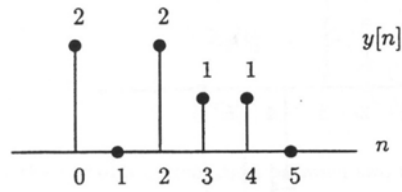
From property 5 in Table 8.2,

$$\text{DFT}\{x_1[((n-4))_8]\} = W_8^{4k} X_1[k]$$

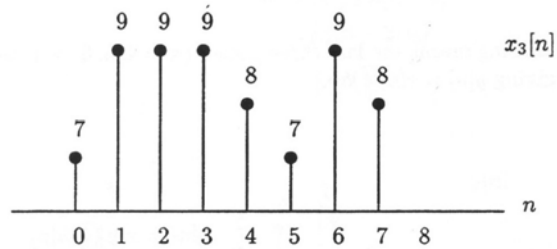
Thus,

$$\begin{aligned} X_2[k] &= W_8^{4k} X_1[k] \\ &= e^{-j\pi k} X_1[k] \\ X_2[k] &= (-1)^k X_1[k] \end{aligned}$$

4. Using the properties of the DFT, we get $y[n] = x[((n-2))_5]$, that is $y[n]$ is equal to $x[n]$ circularly shifted by 2. We get:



5. $x_3[n]$ is the linear convolution of $x_1[n]$ and $x_2[n]$ time-aliased to $N = 8$. Carrying out the 8-point circular convolution, we get:



We thus conclude $x_3[2] = 9$.

6. Looking at the sequences, we see that $x_1[n] * x_2[n]$ is non-zero for $1 \leq n \leq 8$. The smallest N such that $x_1[n] \circledast x_2[n] = x_1[n] * x_2[n]$ is therefore $N = 9$.

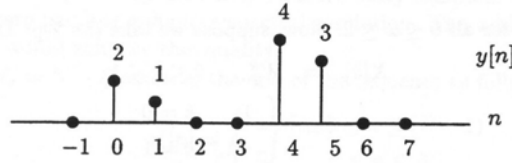
7. (a) When multiplying the DFT of a sequence by a complex exponential, the time-domain signal undergoes a circular shift.

For this case,

$$Y[k] = W_6^{4k} X[k], \quad 0 \leq k \leq 5$$

Therefore,

$$y[n] = x[((n-4))_6], \quad 0 \leq n \leq 5$$



- (b) There are two ways to approach this problem. First, we attempt a solution by brute force.

$$\begin{aligned} X[k] &= 4 + 3W_6^k + 2W_6^{2k} + W_6^{3k}, \quad W_6^k = e^{-j(2\pi k/6)} \text{ and } 0 \leq k \leq 5 \\ W[k] &= \mathcal{Re}\{X[k]\} \\ &= \frac{1}{2} (X[k] + X^*[k]) \\ &= \frac{1}{2} (4 + 3W_6^k + 2W_6^{2k} + W_6^{3k} + 4 + 3W_6^{-k} + 2W_6^{-2k} + W_6^{-3k}) \end{aligned}$$

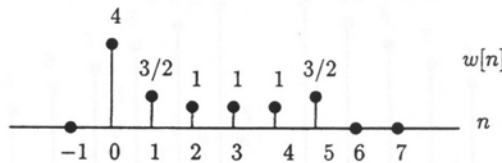
Notice that

$$\begin{aligned} W_N^k &= e^{-j(2\pi k/N)} \\ W_N^{-k} &= e^{j(2\pi k/N)} = e^{-j(2\pi/N)(N-k)} = W_N^{N-k} \\ W[k] &= 4 + \frac{3}{2} [W_6^k + W_6^{6-k}] + [W_6^{2k} + W_6^{6-2k}] + \frac{1}{2} [W_6^{3k} + W_6^{6-3k}], \quad 0 \leq k \leq 5 \end{aligned}$$

So,

$$\begin{aligned} w[n] &= 4\delta[n] + \frac{3}{2} (\delta[n-1] + \delta[n-5]) + \delta[n-2] + \delta[n-4] \\ &\quad + \frac{1}{2} (\delta[n-3] + \delta[n-3]) \\ w[n] &= 4\delta[n] + \frac{3}{2}\delta[n-1] + \delta[n-2] + \delta[n-3] + \delta[n-4] + \frac{3}{2}\delta[n-5], \quad 0 \leq n \leq 5 \end{aligned}$$

Sketching $w[n]$:

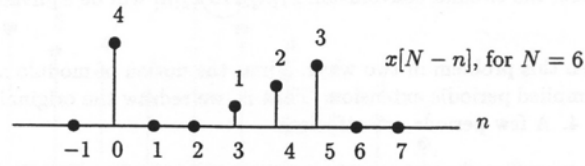


As an alternate approach, suppose we use the properties of the DFT as listed in Table 8.2.

$$\begin{aligned} W[k] &= \mathcal{Re}\{X[k]\} \\ &= \frac{X[k] + X^*[k]}{2} \\ w[n] &= \frac{1}{2} \text{IDFT}\{X[k]\} + \frac{1}{2} \text{IDFT}\{X^*[k]\} \\ &= \frac{1}{2} (x[n] + x^*[((-n))_N]) \end{aligned}$$

For $0 \leq n \leq N - 1$ and $x[n]$ real:

$$w[n] = \frac{1}{2} (x[n] + x[N - n])$$



So, we observe that $w[n]$ results as above.

- (c) The DFT is decimated by two. By taking alternate points of the DFT output, we have half as many points. The influence of this action in the time domain is, as expected, the appearance of aliasing. For the case of decimation by two, we shall find that an additional replica of $x[n]$ surfaces, since the sequence is now periodic with period 3.

From part (b):

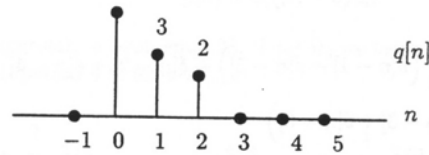
$$X[k] = 4 + 3W_6^k + 2W_6^{2k} + W_6^{3k}, \quad 0 \leq k \leq 5$$

Let $Q[k] = X[2k]$,

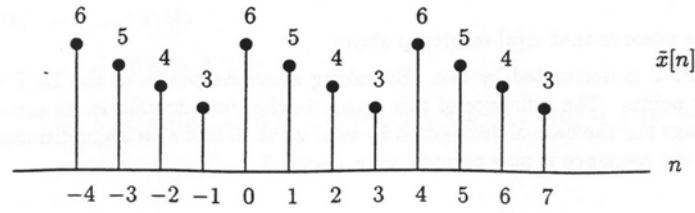
$$Q[k] = 4 + 3W_3^k + 2W_3^{2k} + W_3^{3k}, \quad 0 \leq k \leq 2$$

Noting that $W_3^{3k} = W_3^{0k}$

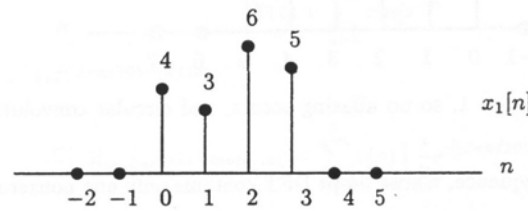
$$q[n] = 5\delta[n] + 3\delta[n-1] + 2\delta[n-2], \quad 0 \leq n \leq 2$$



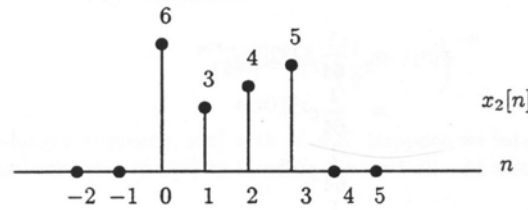
8. We may approach this problem in two ways. First, the notion of modulo arithmetic may be simplified if we utilize the implied periodic extension. That is, we redraw the original signal as if it were periodic with period $N = 4$. A few periods are sufficient:



To obtain $x_1[n] = x[((n-2))_4]$, we shift by two (to the right) and only keep those points which lie in the original domain of the signal (i.e. $0 \leq n \leq 3$):



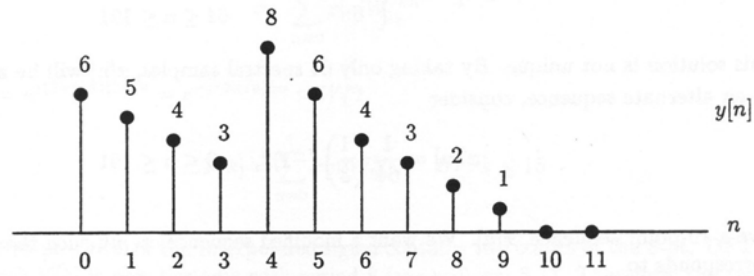
To obtain $x_2[n] = x[((-n))_4]$, we fold the pseudo-periodic version of $x[n]$ over the origin (time-reversal), and again we set all points outside $0 \leq n \leq 3$ equal to zero. Hence,



Note that $x[((0))_4] = x[0]$, etc.

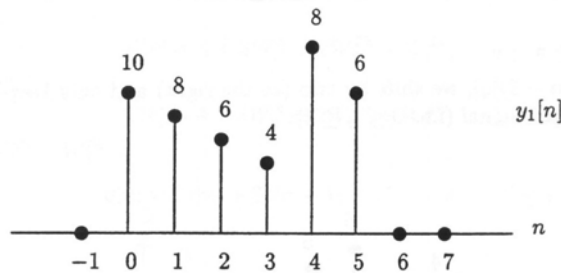
In the second approach, we work with the given signal. The signal is confined to $0 \leq n \leq 3$; therefore, the circular nature must be maintained by picturing the signal on the circumference of a cylinder.

9. Circular convolution equals linear convolution plus aliasing. First, we find $y[n] = x_1[n] * x_2[n]$:



Note that $y[n]$ is a ten point sequence ($N = 6 + 5 - 1$).

- (a) For $N = 6$, the last four non-zero point ($6 \leq n \leq 9$) will alias to the first four points, giving us $y_1[n] = x_1[n] \oplus x_2[n]$



- (b) For $N = 10$, $N \geq 6 + 5 - 1$, so no aliasing occurs, and circular convolution is identical to linear convolution.